

Meritech Sigma-PA Technical & Architecture Manual

Chapter 1: Product Overview

1.1 Introduction to Sigma-PA

Meritech Sigma-PA (Packet Analyzer) is a tool that runs directly on your local machine to monitor and analyze network traffic in real time. It works like a digital detective, capturing and reading data packets as they travel through your network to make sure everything is running smoothly and securely.

Unlike older tools that need to send your data to the cloud to analyze it, Sigma-PA does all its work locally. This keeps your private company information safe inside your own network and avoids slow cloud response times or expensive subscription fees.

1.2 Key Capabilities

Sigma-PA is built to help network engineers and system administrators quickly fix network problems using three main features:

- **No Dropped Data:** It captures network traffic at very high speeds without losing track of a single piece of information.
- **Deep Inspection:** It reads deep inside network packets to identify over 4,000 different applications and protocols (like web traffic, emails, or database queries).
- **Local Storage:** It saves your history in a local database right on the machine, meaning you never have to pay for cloud data processing.

Chapter 2: Hardware Options & How It Connects

2.1 Hardware Tiers

Sigma-PA comes in three physical hardware sizes depending on how much traffic your company handles. Inside each appliance, a high-performance network capture card routes data using optimized internal links straight to a fast local storage array. The main system processor handles all deep packet analysis locally on the machine.

- **Sigma-PA 2000 (For Small Offices):** A compact device built for smaller branch offices, handling up to 10 Gbps of traffic.
- **Sigma-PA 5000 (For Core Data Centers):** A standard enterprise device that connects directly to main company switches, handling up to 100 Gbps.
- **Sigma-PA 9000 (For Huge Topologies):** A heavy-duty system made for massive data hubs, capable of reading up to 400 Gbps.

2.2 How It Hooks Up to Your Network

You can connect Sigma-PA to your network in two different ways:

1. **Passive Mode (Listening Only):** The device hooks up to a copy port on your switch. It listens to a duplicate stream of traffic without getting in the way. If the device turns off, your network keeps running completely fine.

2. **Inline Mode (Bridge Mode):** The device sits directly in the middle of the cable path. This allows it to block bad traffic in real time. If the device crashes, a built-in safety switch connects the cables internally so your internet doesn't go down.

Chapter 3: The Internal Software Components

Sigma-PA splits its work into three main software layers:

3.1 Packet Capture Core (PCC)

This layer acts as the collector. It bypasses the slow standard operating system pathways to grab data straight from the network card. It slaps a highly accurate time stamp on every single packet so you know exactly when it arrived.

3.2 Protocol Dissection Engine (PDE)

This layer acts as the translator. It opens up the captured packets layer by layer.

- First, it unpacks Layer 2 (the Ethernet frame layout).
- Second, it reads Layer 3 (the IP addressing data).
- Third, it tracks Layer 4 (the TCP connection management).
- Finally, it decodes Layer 7 (the actual application text or files like HTTP, DNS, or SQL).

3.3 Storage Indexer & Metadata Layer

Writing raw packet files straight to a hard drive can slow down a computer. To fix this, Sigma-PA extracts a tiny summary of each chat (like who sent it, where it went, and how large it was) and saves that summary to a fast database index. The heavy raw files are compressed and saved separately in the background.

Chapter 4: AI & RAG Integration Layout

4.1 Local Vector Storage Friendly

Sigma-PA is built to work with modern AI setups. Instead of saving system logs in hard-to-read text jumbles, it formats logs into clean, structured blocks. This makes it easy for local vector databases (like a local Chroma database) to chunk and store the information without throwing format errors.

4.2 Keeping Data Private

When an engineer uses a local AI chatbot to fix a network bug, Sigma-PA provides the local context blocks. Because everything happens within your private network, sensitive details like your internal IP addresses are never sent to external web APIs or cloud networks.

Chapter 5: Advanced Inspection Features

5.1 Deep Packet Inspection (DPI)

Standard network tools only look at port numbers. If malicious traffic disguises itself on a standard web port like 443, older tools get fooled. Sigma-PA looks inside the actual message payload to verify exactly what kind of application is running, easily separating standard web browsing from hidden files or database actions.

5.2 Handling Encrypted Data (TLS 1.3)

Most modern network traffic is encrypted. Sigma-PA can analyze the initial unencrypted handshake exchange to spot fake security certificates or suspicious connection setups without needing to decrypt the payload. For extreme security cases, it can connect with secure hardware modules to safely decrypt, check, and instantly erase data from memory to maintain strict user privacy.

Chapter 6: Step-by-Step Test Drive

Follow these simple commands in your terminal to check if your Sigma-PA appliance is running and capturing traffic properly:

Step 1: Check Available Links

Run this command to see which ports are connected:

```
sigmapa-admin --list-interfaces
```

Step 2: Start a 60-Second Test Capture

Tell the system to watch your web traffic for one minute and save it to a test file:

```
sigmapa-core --interface cap0 --output test_run.pcap --filter "tcp port 80 or tcp port 443" --duration 60s
```

Step 3: View the Results

Check the index summary to make sure the data was recorded cleanly:

```
sigmapa-query --input test_run.pcap --summary-stats
```

Chapter 7: The Difference Between Sigma-LA and Sigma-PA

While both devices are made by Meritech, they handle two completely different jobs:

- **Sigma-LA (Link Analyzer):** Looks at the physical hardware wires and links (Layers 1-2). It measures signal strengths, laser light levels, and hardware cable errors. Think of it as checking if the road is paved smoothly and the traffic lights work.
- **Sigma-PA (Packet Analyzer):** Looks at the actual data contents and apps (Layers 3-7). It measures application names, text strings, and raw files. Think of it as looking inside the delivery trucks to check what goods they are carrying.

Deploying **Sigma-LA** ensures your physical network pipes aren't broken, while deploying **Sigma-PA** ensures the data flowing inside those pipes is accurate, safe, and clean.

Chapter 8: Why Sigma-PA is Better Than Competitors

1. **100% Offline Operations:** You don't need an internet connection to use it. Your network data stays private and safe within your walls.
2. **No Slowdowns:** By grabbing data directly from the network hardware memory, it never bogs down the operating system, even during massive traffic spikes.

3. **Instant Search Responses:** Because it extracts and indexes small metadata summaries in real time, you can find a specific target packet in seconds instead of waiting hours for huge files to load.

Chapter 9: Real-World Troubleshooting Examples

9.1 Solving the "Is it the network or the app?" Argument

When a company application slows down, different teams usually blame each other. Sigma-PA settles this by showing exact numbers. By looking at the time stamps, it can prove if a network transfer took less than 2 milliseconds but the application server took over 4 seconds to respond. This instantly proves the network is fine and the app code needs fixing.

9.2 Spotting Internal Security Threats

Firewalls block external hackers, but they can miss a compromised internal computer sending sensitive data out. Sigma-PA constantly audits internal traffic. If an internal server suddenly starts sending massive files over a standard web port, Sigma-PA automatically flags the action and captures the relevant packets for review.

Chapter 10: Simple Storage Calculation

To figure out how much hard drive space you need to save your network history, use this simple formula:

$$\text{Storage Needed} = \text{Network Traffic Speed} \times \text{How long you want to keep it}$$

For example, if you run a continuous **10 Gbps** network link, it generates **1.25 Gigabytes of data every second**. To save every single packet completely for a full **24-hour day**, you will need **108 Terabytes** of storage space.

Chapter 11: Security & Compliance Settings

11.1 Isolating Management Access

To prevent hackers from messing with your analyzer, Sigma-PA keeps its management tools completely separated from its capture tools:

- **Capture Ports (cap0, cap1):** These ports have no network IP configuration. They are completely invisible ("stealth mode") and cannot be pinged or attacked from the outside.
- **Management Port (eth0):** This is the only port that allows you to log into the settings panel, and it can be hidden away on a private, locked-down administration network.

11.2 Privacy Safeguards

If your company handles sensitive customer data like credit cards or medical histories, Sigma-PA can automatically mask those numbers out with blank characters before writing the packet to the local disk, keeping your network compliant with privacy laws.

Chapter 12: Simple Python Automation API

Sigma-PA includes a built-in REST API that responds with standard JSON code, making it incredibly simple to write basic Python scripts to check on your network stats. Here is a short, readable example script to search your device for active web traffic records:

Python

```
import requests
```

```
import json
```

```
APPLIANCE_IP = "10.0.10.25"
```

```
API_KEY = "ALPHA_SIGMA_PA_SECURE_TOKEN"
```

```
headers = {
```

```
    "Authorization": f"Bearer {API_KEY}",
```

```
    "Content-Type": "application/json"
```

```
}
```

```
query = {
```

```
    "target_protocol": "HTTP/2",
```

```
    "limit": 3
```

```
}
```

```
try:
```

```
    url = f"http://{APPLIANCE_IP}/api/v1/analytics/streams"
```

```
    response = requests.post(url, headers=headers, data=json.dumps(query))
```

```
    if response.status_code == 200:
```

```
        data_streams = response.json()["results"]
```

```
        print("Found active network connections:")
```

```
        for stream in data_streams:
```

```
            print(f"- From: {stream['src_ip']} -> To: {stream['dest_ip']} | Size: {stream['total_bytes']} bytes")
```

```
    except Exception as error:
```

```
        print(f"Could not connect to the Sigma-PA device: {error}")
```

Chapter 13: Technical Troubleshooting

- **Problem:** The capture port status says "DOWN".

- *The Fix:* Check your fiber patch cables to make sure the send (TX) and receive (RX) lines aren't switched, and verify that your network switch port mirroring feature is turned on.
- **Problem:** The system logs say "Frame Drops".
 - *The Fix:* Your storage drives are getting overwhelmed by too much data at once. Adjust your input filters to ignore safe, high-volume internal company backups to free up processing space.

Chapter 14: Daily Maintenance Lifecycle

Every night at 2:00 AM, the Sigma-PA software runs an automated self-cleanup loop. It checks the health of your local storage drives, purges old packet history rows that are past their retention date expiration, and updates its local search summaries so the app stays fast and responsive when you use it the next day.

Chapter 15: Frequently Asked Questions (FAQ)

Q1: Will adding Sigma-PA slow down my internet or corporate network speeds?

No. When used in its standard Passive Mode, the device only reads a duplicate copy of your traffic sent from a switch port. It doesn't touch the original traffic, meaning it has zero impact on your active network speeds.

Q2: Can I use this for virtual networks like AWS or Microsoft Azure?

Yes! Meritech offers a Virtual Edition (VE) version of this software. You can run it inside your cloud environment as a virtual machine, and it will capture and read your virtual cloud traffic just like the physical hardware box does in a data center.

Q3: How do I update the application list if the device is kept completely offline?

You download update files from our secure website using a regular internet-connected computer. Then, you move the file onto a local administrative laptop and upload it directly into the appliance's management port using a local cable connection.

Q4: Can I send specific alert logs to external dashboard systems?

Yes. Even though Sigma-PA does its storage and processing locally for security, you can configure it to securely stream text summaries or error alerts to external security monitoring platforms like Splunk or Elasticsearch.

Q5: What happens if the device fills up its hard drive storage space entirely?

The device uses a rolling retention rule. When the hard drive hits capacity, it automatically drops the oldest saved days of history to make room for new incoming packets, keeping your system running smoothly without manually erasing anything.